

Algorithm	Area where used	How it is used	Why it is used
K-Means	 Customer segmentation	Groups customers based on things like spending and purchase behavior	To find natural customer groups
Hierarchical Clustering	 Biology, customer analysis	Creates groups step-by-step and forms a hierarchy/tree	Useful when we want to see relationships between groups
DBSCAN	 Location/GPS data, anomaly detection	Groups points that are closely packed and identifies isolated points as noise	Finds irregular-shaped clusters and outliers
PAM (K-Medoids)	 Customer segmentation	Groups data around actual representative data points called medoids	More robust to outliers than K-Means
PCA	 Data preprocessing, visualization	Converts many features into fewer important principal components	Reduces dimensions while retaining important information
Association Rule Learning	 Market basket analysis	Finds items that are frequently purchased together	To discover purchase patterns